



COCKRELL SCHOOL OF ENGINEERING
McKetta Department of Chemical Engineering

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CHE 322: Thermodynamics

Fall 2026 #17030

CLASS TIME: 11:00 am – 12:30 pm Tue/Thur in CPE 2.216

RECITATION: 1:00 pm – 3:00 pm Wed in PMA 7.104

Instructor: Kent Zheng

Office hours: Tuesday 5-6 pm (CPE 3.404)

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Tutor: Vishal Subramanian

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Grader: Alex Kretzschmar & Huyen Nguyen

Required Texts:

The course will very closely follow these two textbooks.

- Koretsky *"Engineering and Chemical Thermodynamics" 2nd Ed. (Primary)*
- Gaskell & Laughlin *"Introduction to the Thermodynamics of Materials" 7th Ed.*

Course Description: Thermodynamics relates work, heat, temperature and states of matter to each other. From a surprisingly small set of empirically based laws, an enormous amount of information about the relationships among equilibrium parameters for a system can be deduced. This information can then be applied to physical, chemical and biological systems including engine design, materials processing and cellular processes. Amazingly, thermodynamics is independent of any molecular model of matter, but molecular interpretations of various aspects of the subject (e.g., entropy and temperature) will be discussed in the course to broaden understanding. The focus of this course is the further development of thermodynamics based on the Ch353 Physical Chemistry pre-requisite and the application of the subject to practical systems.

Goals: The objective of this course is to introduce students to the principles of thermodynamics as they apply to physical and chemical processes.

Impact on Subsequent Courses in Curriculum:

Thermodynamic properties, phase equilibrium and chemical reaction equilibrium play an important role throughout chemical engineering, e.g. in ChE 360, ChE 363, ChE 372 and ChE 473K. Thermodynamics is one of the main pillars of chemical engineering; others include transport phenomena and reaction kinetics.

Course procedure

- Attendance in lectures and discussions is strongly encouraged. I personally do not see a realistic way to excel in, or even pass, this course with limited in-person participation. If you attend class, please arrive on time so that you do not miss the setup and motivation for that day's lecture.
 - Lecture hours will be used to introduce new material, provide detailed examples, conduct some quizzes, conduct three hourly exams.
 - Canvas will be used to post all course materials, assignment grades and any announcements. Please check Canvas on a daily basis.

Grading

Homework	5%
Quizzes	5%
Midterm (x3)	20% each
Final	30%
Total	100%

Final grades will not be curved, by default. (*subject to change)

GRADE SCALE: >88% (A); 85-88 (A-); 82-85 (B+); 73-82% (B); 70-73 (B-); 67-70 (C+); 58-67% (C); 55-58 (C-); 50-55% (D); <50% (F)

Critical & meaningful AI use is “allowed” and “encouraged” throughout the course, but you are solely on your own in the exams (90% of final grade, closed-book, ¼ single-page equation sheet allowed).

Students with disabilities The University of Texas at Austin provides upon request appropriate academic adjustments for qualified students with disabilities. For more information, contact the Office of the Dean of Students at 471-6259, 471-4641 TTY or the College of Engineering Director of Students with Disabilities at 471-4321. Division of Diversity and Community Engagement, Services for Students with Disabilities, 512-471-6259, <http://www.utexas.edu/diversity/ddce/ssd/>

Academic dishonesty UT Honor Code (or statement of ethics) and an explanation or example of what constitutes plagiarism (Link to University Honor Code: <http://catalog.utexas.edu/general-information/the-university/#universitycodeofconduct>)

Religious holidays By UT Austin policy, you must notify the instructor of your pending absence at least fourteen days prior to the date of observance of a religious holy day. If

you must miss a class, an examination, a work assignment, or a project in order to observe a religious holy day, you will be given an opportunity to complete the missed work within a reasonable time after the absence.”)

Emergency evacuations Recommendations regarding emergency evacuation are available from the Office of Campus Safety and Security, 512-471-5767, <http://www.utexas.edu/safety/>. For CPE 2.220 specifically, in the event of an emergency, we egress from CPE and assemble on the south side of the pedestrian bridge until the all-clear notification. You can sign up to receive emergency text alerts here: <https://utdirect.utexas.edu/apps/csas/text/>.

Course Content Outline

(*subject to change)

Topic 0. Context and Language of Thermodynamics (Koretsky ch1)

Topic 1. The First Law of Thermodynamics (ch2)

Topic 2. The Second Law of Thermodynamics (ch3)

Topic 3. Equation of State (ch4)

Topic 4. The Thermodynamic Web (ch5)

Topic 5. Phase Equilibria (ch6~8)

Topic 6. Other Applications (ch9)

6.1 Chemical Reactions

6.2 Electrochemical Reactions